TUMELO MOENG

Exercise 2

Question 1:

```
SELECT DISTINCT department
FROM            student;
```

department
IT
HR
FINANCE

Question 2:

```
SELECT   department,
         AVG (age) As average_age
FROM     students
GROUP BY department;
```

department	average_age
IT	20
HR	22
Finance	23

Question 3:

```
SELECT   department,
         COUNT(*) AS student_count
FROM     students
GROUP BY department
HAVING   COUNT(*) > 1;
```

department	student_count
IT	2
HR	2

Question 4:

```
SELECT   *
FROM     students
WHERE    age is BETWEEN 21 AND 23;
```

Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Question 5:

```
SELECT   *
FROM     students
WHERE    department = 'IT' OR 'HR' >21
```

student-id	name	age	departmene
2	Bob	22	HR
~~4~~ 3	~~Diana~~ Charlie	21	IT
5	Eve	22	HR

Question 6:

Courses

course-id	course-name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Eaccel	Finance	2
105	Statistics	HR	2

```
SELECT      department,
            ~~Credit~~ Sum(credits) ~~Count(*)~~ AS total_credit
FROM        courses
GROUP BY    department
Having      Sum(credits) ~~COUNT(*)~~ > 5
```

department	total-credit
IT	11

Question 7:

~~SELECT * FROM courses WHERE~~

```
SELECT      *
FROM        courses
WHERE       course Not have 4 credits
```

course-id	course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8:

```
SELECT      course_id,
            course_name,
            credits
FROM        courses
ORDER BY    credits DESC
limit 3;
```

course-id	course-name	credits
103	Data Science	4
102	Python	4
101	SQL Basics	3

Question 9:

enrollments

enrollment-id	student-id	course-id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

1)

```
SELECT    Max(Grade) AS max_grade
FROM      enrollments
```

Max_grade
90

```
SELECT    MIN(grade) AS min_grade
FROM      enrollments
```

Min-grade
78

2)

```
SELECT    Average(grade) AS Avg-grade
FROM      enrollments
```

Avg_grade
84,6

Question 10:

```
SELECT    course-id,
          COUNT(*) AS enrollment-count
FROM      enrollments
GROUP BY  course-id;
```

Course-id	enrollment-count
101	1
102	2
103	3
104	4
105	5

Question 11:

salaries

employee-id	name	department	salary	bonus
1	Tom	IT	60 000	5 000
2	Jerry	HR	55 000	4 000
3	Spike	Finance	70 000	6 000
4	Tyke	IT	62 000	5 500
5	Butch	HR	54 000	3 500

```
SELECT    department,
          SUM(salary) AS total-salary,
          SUM(Bonus) AS total-bonus
FROM      salaries
GROUP BY  department
```

department	total-salary	total-bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

Question 12:

```
SELECT    department,
          Average(salary) AS Avg-salary
FROM      salaries
GROUP BY  department
HAVING    Avg-salary >55 000
```

department	Avg-salary
IT	61 000

Question 13:

```
SELECT    employee-id
          name,
          salary,
          bonus.
FROM      salaries
WHERE     ~~salary + bonus > 60 000~~
          salary + bo
```

```
SELECT    employee-id
          name,
          salary,
          bonus
FROM      salaries
WHERE     SUM(salary + bonus) AS total-compensation > 60 000
```

employee-id	name	salary	bonus	total-compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tylce	62 000	5 500	67 500

Projects

project-id	project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

Question 14:

```
SELECT    department
          SUM(budget) AS total_budget,
          Average(budget) AS avg-budget
FROM      projects
GROUP BY  department
Having    average-budget > 70 000
```

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

Question 15 :

```
SELECT    *
          project-id,
          project-name,
          department,
          budget
FROM      projects
WHERE     budget BETWEEN 50 000 AND 120 000 NOT Marketing
```

project-id	project-name	department	budget
1	A.I App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000